

Muhammad Hassan

AI/ML Engineer

Computer Science undergraduate specializing in AI/ML engineering with hands-on experience building LLM-powered applications, RAG pipelines, and end-to-end machine learning systems. Proficient in Python, LangChain, ChromaDB, OpenAI API, and Scikit-learn (classification, regression, and clustering algorithms). Developed and deployed multiple real-world ML projects including fraud detection, predictive maintenance, and AI research assistants. Actively seeking an AI/ML internship to apply and expand practical skills.

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EDUCATION

Bachelor of Science in Computer Science

University of Central Punjab

Oct 2023 - Jul 2027 | Lahore, Pakistan

PROJECTS

Credit Card Fraud Detection

Website

Developed a fraud detection model using **Random Forest** and **SMOTE** on highly imbalanced transaction data, achieving **92% precision** and **81.2% F1-score**. Deployed as a Streamlit web application for real-time and batch fraud prediction.

Heart Disease Prediction

Website

Developed a heart disease risk prediction system comparing four machine learning models (**Random Forest**, **Logistic Regression**, **Naive Bayes**, and **KNN**), with **Random Forest** achieving **87% accuracy**. Deployed as a Streamlit application for real-time risk assessment and probability-based predictions.

AI Multi-Agent Research Assistant:

Website

Built a multi-agent AI research system using **LangChain**, **Cerebras LLM**, **Exa API**, and **Streamlit** to perform real-time web research and generate source-backed reports. Implemented RAG and AI-driven query expansion for deeper analysis and executive-level summaries.

Machine Failure Prediction

Website

Built a machine failure prediction system using industrial sensor data and compared multiple ML models to classify failure types.

SKILLS

Python

Langchain

Scikit-Learn

Pandas

Numpy

Matplotlib

C++

SQL

RAG (Retrieval-Augmented Generation).

LLMs

Vector Databases (Pinecone, Faiss, Chroma DB).

APIs

Feature Engineering

Data Preprocessing

EDA

Data Visualization

CERTIFICATES

Supervised MachineLearning: Regression and Classification

Coursera

AWS Academy Graduate

AWS Cloud Foundations

Zero to Hero in LangChain:Build GenAI Apps

Udemy

Python Programming

Udemy

LANGUAGES

Urdu

Native or Bilingual
Proficiency

English

Professional Working
Proficiency

TOOLS & SOFT SKILL

GitHub

Jupyter

AWS

Claude

Cursor